Mohammad Pouya Toroghi

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Education

M.Sc. in Computer Engineering — Sharif University of Technology
GPA: 18.38/20 — Sept. 2024 - Present
Thesis: Detecting Backdoor Attacks in Large Language Models. — Advisor: Prof. Mohammad Hossein Rohban

B.Sc. in Electrical Engineering — Sharif University of Technology
GPA: 18.47/20 — June 2021 - June. 2024
Thesis: Machine Unlearning for Neural Networks. — Advisors: Prof. Arash Amini, Prof. Stefano Rini

B.Sc. in Electrical Engineering — University of Tehran
GPA: 17.41/20 — Sept. 2020 - June 2021
Transferred to Sharif University to continue my undergraduate education.

Research Interests

Trustworthy ML, LLMs, Reinforcement Learning, Neuro-Symbolic AI, Explainable AI, Machine Unlearning

Research Experience

- **LLM Backdoor Detection | RIML Lab, Sharif University of Technology** — Sept. 2024 - Present
  Advisor: Prof. Mohammad Hossein Rohban

  Conducting M.Sc. research on detecting backdoored large language models with full white-box access, without assuming a known trigger distribution. Developing a search-based detection approach alongside partial estimation of the trigger distribution under explicit assumptions. The work investigates how white-box access can guide the search for candidate triggers and how the resulting information can support the identification of backdoored models. Evaluating model responses under controlled attack settings to assess detection performance, the usefulness of the estimated trigger information, and the limitations imposed by the underlying assumptions.

- **Calibration-Aware Fine-Tuning | Remote Research Project with CUHK** — Dec. 2025 - Present
  Advisor: Prof. Amin Gohari

  Investigating semantic calibration in fine-tuned large language models for long-form generation, focusing on overconfidence under distribution shift and limited fine-tuning data. Contributed to problem formulation and am developing training-time methods that use in-domain supervision to align confidence with empirical response correctness while preserving generation quality and limiting inference overhead. Implementing training and evaluation pipelines, comparing relevant baselines, and analyzing calibration-accuracy and inference-cost trade-offs. The methodology and manuscript are being refined for submission to ICML, with an emphasis on reliable uncertainty estimation without extensive inference-time sampling.

- **Machine Unlearning | Sharif University of Technology** — Sept. 2023 - Sept. 2024
  Advisors: Prof. Arash Amini *&* Stefano Rini

  Conducted B.Sc. thesis research on machine unlearning for neural networks, investigating the removal of training-data influence and its implications for model utility, privacy, and security. Implemented and compared existing unlearning algorithms across multiple settings and developed and evaluated a federated unlearning approach for distributed training. Designed an empirical evaluation protocol combining retained-task performance, forgetting effectiveness, membership-inference attack success, and poisoning-attack evaluations before and after unlearning to assess the effectiveness and limitations of the methods.

Honors and Awards

- Admitted to Sharif University of Technology's M.Sc. program in Computer Engineering through the Exceptional Talents Initiative, with exemption from the entrance examination — June 2024
- Ranked **18th** in my class of 137 Electrical Engineering students at Sharif University of Technology. — June 2024
- Ranked **331st** nationally among approximately 200,000 candidates in the Mathematics-Physics track of Iran's National University Entrance Examination — Sept. 2020

Teaching Experience

- Teaching Assistant for Deep Reinforcement Learning
  - Head Teaching Assistant, Instructor: Prof. Mohammad Hossein Rohban. — Spring 2026
- Teaching Assistant for Deep Generative Models
  - Project Design, Instructor: Prof. Sajjad Amini. — Fall 2024
- Teaching Assistant for Signals & Systems
  - Homework, and Exam Design, and TA Sessions, Instructor: Prof. Hossein Sameti. — Fall 2024
  - Homework Design, Instructor: Prof. Babak Hossein Khalaj. — Spring 2023
- Teaching Assistant for Communication Systems
  - Homework Design, Instructor: Prof. Mohammad Hadi. — Fall 2024
  - Homework Design, Instructor: Prof. Fereidoon Behnia. — Fall 2024
  - Head Teaching Assistant, Instructor: Prof. Mohammad Mahdi Mojahedian. — Fall 2023
- Teaching Assistant for Circuit Theory
  - Homework Design, Instructor: Prof. Emad Fatemizadeh. — Fall 2024
- Teaching Assistant for Machine Learning
  - Homework, and Project Design, Instructor: Prof. Rouhollah Amiri. — Fall 2024
  - Homework, and Project Design, Instructor: Prof. Mohammad Bagher Shamsollahi. — Fall 2023, Spring 2024
- Teaching Assistant for Digital Communications
  - Homework Design, Instructor: Prof. Fereidoon Behnia. — Spring 2024
- Teaching Assistant for Linear Algebra
  - Homework Design, Instructor: Prof. Alireza Farhadi. — Spring 2024
- Teaching Assistant for Engineering Mathematics
  - Homework Design, Instructor: Prof. Davod Poreh. — Spring 2023

Standardized Test

TOEFL: Overall Score: **112** (Reading: 30, Listening: 30, Speaking: 25, Writing: 27) — Nov. 18, 2023

Selected Projects

- **LLM Reasoning, Tool-Using Agents & Neuro-Symbolic AI** — Python, PyTorch, Hugging Face Transformers, TRL, vLLM, NetworkX, SymPy

  Developed systems for multi-step reasoning, program synthesis, and tool-augmented LLM inference. Implemented neuro-symbolic pipelines using seq2seq LSTM and Transformer architectures for program generation and execution-based evaluation. Built symbolic regression models using neural-network and Transformer architectures. Implemented inference-time reasoning and search methods spanning chain-of-thought prompting, best-of-N sampling, beam search, self-refinement, Tree of Thoughts, A* search, and Monte Carlo tree search. Built retrieval-augmented and graph-based reasoning pipelines integrating entity graph construction, clustering, and structured LLM evaluation.

- **Reinforcement Learning Algorithms & Systems** — Python, PyTorch, Gymnasium, NumPy

  Integrated theoretical study with practical implementation and empirical evaluation of reinforcement learning algorithms spanning Monte Carlo methods; temporal-difference learning (Q-learning, SARSA); policy-gradient methods (REINFORCE, PPO); value-based deep RL (DQN variants); and off-policy actor-critic methods (SAC, DDPG). Extended these implementations to imitation learning, offline RL, multi-agent RL, hierarchical reinforcement learning, and exploration methods.

- **Deep Learning & Generative Modeling Systems** — Python, PyTorch, TensorFlow, Hugging Face Transformers, Diffusers

  Designed and implemented deep learning models for vision, language, and generative modeling, including CNN classifiers, segmentation networks, sequence models for temporal prediction, and Transformer-based language models. Built parameter-efficient fine-tuning workflows using LoRA and prefix tuning, and experimented with variational autoencoders (VAEs), DDPMs, and latent diffusion systems. Experimentally investigated representation learning and visualized attention in vision-language models.

- **Classical AI, Search, Constraint Solving & Probabilistic Models** `Python, NumPy, OR-Tools, NetworkX`

  Implemented classical AI algorithms for graph search (DFS, BFS, A*), local search (hill climbing, simulated annealing), and adversarial search (minimax with alpha-beta pruning). Developed constraint-solving pipelines using backtracking, AC-3/AC-4, and CP-SAT. Implemented probabilistic inference in Bayesian networks using variable elimination and sampling, supervised-learning methods for regression and classification, including decision trees, and sequence models using RNN and GRU architectures.

- **Machine Learning System Design (Medical Imaging & Representation Learning)** `Python, PyTorch, NumPy, scikit-learn`

  Designed and implemented image representation learning and restoration pipelines using autoencoders for image colorization and super-resolution, alongside PCA-based reconstruction and neural autoencoders for denoising. Applied Gaussian mixture models fitted using expectation-maximization to medical image analysis. Assessed the resulting models in terms of image reconstruction quality and performance on clustering-based segmentation tasks.

- **Speech Recognition System** `Python, PyTorch, torchaudio`

  Developed a Persian automatic speech recognition system using a Conformer-Transducer architecture. Built audio-normalization and preprocessing pipelines, trained on segmented speech data, and evaluated recognition performance across speakers and varying acoustic conditions.

- **Large-Scale NLP & Data Systems** `Python, pandas, NumPy, scikit-learn, PyTorch`

  Built large-scale text analysis pipelines for social media, implementing sentiment classification pipelines using bag-of-words and TF-IDF representations with classical classifiers, alongside Transformer-based models. Developed data collection and web-scraping pipelines with integrated database storage to support large-scale behavioral and trend analysis.

Skills

Languages: Persian: native; English: proficient; Arabic: reading proficiency; German and French: beginner.

Programming & Systems: Python, C++, Java, MATLAB; familiarity with C, Bash, Verilog, Assembly, Go, and HTML.

Additional LLM Methods: QLoRA fine-tuning, GraphRAG, ReAct agents, and PPO-based LLM post-training.